

SOUTH BASS ISLAND, OH, USA

WMO: 994340

Lat: 41.6289N

Lon: 82.8413W

Elev: 580

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.8	(c) 9.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 42.2	(k) 25.5	(l) 38.9	(m) 24.8	(n) 16.7	(o) 230	(p) 0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.7	(c) 84.9	(d) N/A	(e) 82.7	(f) N/A	(g) 80.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.8	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
39.7	34.1	29.5	1.4	89.9	5.8	3.0	-2.8	92.0	-6.2	93.8	-9.4	95.5	-13.6	97.7
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	26.3	27.0	35.5	47.7	58.9	69.6	74.7	73.8	67.6	55.5	43.3	33.2
	DBStd	18.80	10.06	9.40	8.73	7.68	7.29	5.61	4.47	4.17	5.85	7.13	8.00	7.96
	HDD50	2748	735	644	455	131	8	0	0	0	0	25	227	523
	HDD65	5918	1200	1064	913	519	219	20	1	1	39	305	650	987
	CDD50	3195	0	0	7	63	283	587	765	738	527	197	26	2
	CDD65	890	0	0	0	1	29	157	301	274	116	12	0	0
	CDH74	4031	0	0	0	2	82	563	1721	1300	341	22	0	0
	CDH80	580	0	0	0	0	12	78	286	160	43	1	0	0
Wind	WSAvg	13.7	17.0	16.2	14.5	15.6	12.7	10.5	9.8	9.6	11.0	14.4	16.4	16.5
Precipitation	PrecAvg	37.6	2.5	2.2	2.6	3.6	3.7	3.9	3.7	3.5	3.5	2.8	2.8	2.8
	PrecMax	45.7	5.6	5.3	5.4	5.4	7.2	6.3	8.6	8.5	8.1	4.7	5.5	7.2
	PrecMin	26.6	0.7	0.6	1.1	2.0	1.2	1.3	1.6	0.8	1.2	0.7	0.8	1.2
	PrecStd	4.9	1.2	1.2	1.1	1.0	1.7	1.4	1.6	1.8	1.6	0.9	1.0	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	52.5	63.3	72.4	81.5	85.4	88.3	86.9	84.9	77.1	64.3	56.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	49.4	46.2	57.2	67.4	75.7	82.2	85.3	83.8	80.4	72.3	60.2	50.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	44.6	41.8	52.5	63.7	72.4	79.7	83.2	81.6	77.6	68.9	57.3	46.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	39.0	37.8	47.5	59.5	69.2	77.2	81.2	79.8	75.5	65.6	54.4	43.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.2	10.4	11.3	10.5	9.9	9.7	9.0	9.3	9.2	8.7	7.8
		MCDBR	14.6	14.3	17.5	18.1	14.2	12.4	11.5	10.5	11.9	12.8	13.2	12.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.309	0.333	0.359	0.400	0.436	0.448	0.442	0.433	0.387	0.365	0.326	0.316
	taud		2.394	2.305	2.314	2.251	2.192	2.208	2.245	2.257	2.362	2.403	2.475	2.419
	Ebn at Noon		266	275	281	276	268	264	264	263	268	261	258	252
	Edh at Noon		26	33	36	42	45	45	43	41	35	30	24	23
All-Sky Solar Radiation	RadAvg		456	725	1090	1459	1722	1919	1929	1692	1366	891	562	382
	RadStd		47	97	97	115	145	128	107	64	116	70	80	39

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	N/A

Nomenclature: See separate page